

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

25

Q1

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The function f is defined by $fx = -3x + 60$. What is the value of fx when $x = -8$?

A. 49

B. 52

C. 57

D. 84

Q2

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For the linear function f , $f0 = 17$ and $f1 = 17$. Which equation defines f ?

A. $fx = \frac{1}{17}$ B. $fx = 1$ C. $fx = 17$ D. $fx = 34$

Q3

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$$d = 16t$$

The given equation represents the distance d , in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of 16 in this context?

- A. The object moved a total of 16 inches.
- B. The object moved a total of $16t$ inches.
- C. The object is moving at a rate of 16 inches per second.
- D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

Q4

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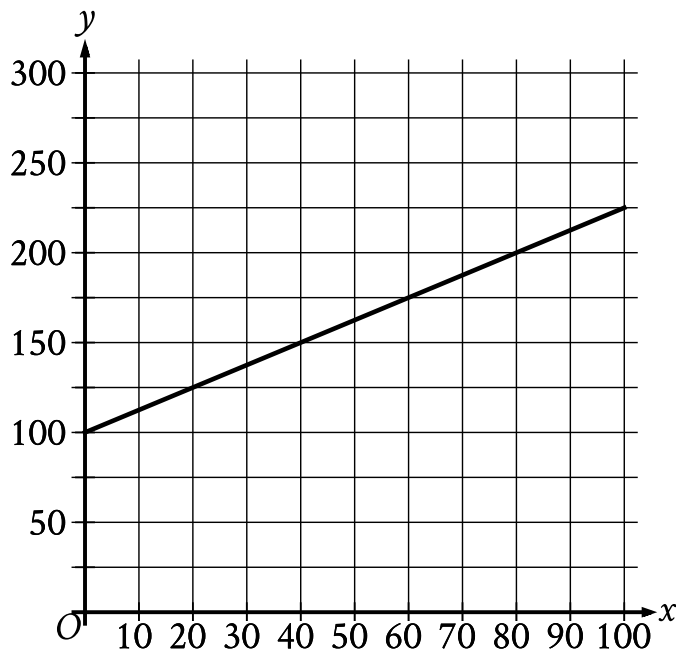
The function g is defined by $g(x) = -x + 8$. What is the value of $g(0)$?

- A. -8
- B. 0
- C. 4
- D. 8

Josie purchased a pass that cost \$ 19.50 to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost \$ 62.00 total. What was the cost of one sticker?

- A. \$ 1.95
- B. \$ 4.25
- C. \$ 6.20
- D. \$ 8.15

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (0 comma 100)
 - (60 comma 175)
 - (100 comma 225)

What is the cost, in dollars, for the manufacturer to make 60 rings?

- A. 100
- B. 125
- C. 175

D. 225

Q7

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The function f is defined by $fx = 5x + 8$. For what value of x does $fx = 58$?

- A. 10
- B. 13
- C. 50
- D. 298

Q8

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$$fx = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

- A. The tree will be measured each year for 4 years.
- B. The tree is estimated to grow to a maximum height of 4 feet.
- C. The estimated height of the tree increased by 4 feet each year.
- D. The estimated height of the tree was 4 feet when it was first measured.

Q9

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The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h , in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

A. $h = 8s + 15$

B. $h = 15s + \frac{335}{8}$

C. $h = 8s + \frac{335}{15}$

D. $h = 15s + 8$

Q10

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Robert rented a truck to transport materials he purchased from a hardware store. He was charged an initial fee of \$20.00 plus an additional \$0.70 per mile driven. If the truck was driven 38 miles, what was the total amount Robert was charged?

A. \$46.60

B. \$52.90

C. \$66.90

D. \$86.50